NOVA IMS INFORMATION MANAGEMENT SCHOOL

DATA MINING PROJECT – 2026/2027

ACDC Coworking

Customer Segmentation

Deliverable 1 | Exploratory Data Analysis - Discussion & Key Findings

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Business & Data Understanding

BUSINESS PROBLEM:

Customers are becoming more selective about where they spend their money, so services and marketing need to match their needs. Treating members as one homogeneous group fails to address their diverse needs

BUSINESS OBJECTIVES:

Understand member differences. Use segmentation to target marketing and tailor services and benefits, improving member satisfaction, retention and revenue

Data objectives: assess missing data, inconsistencies and outliers. Identify and engineer useful features. Develop evidence based segment hypotheses

DATA SOURCE: ACDC membership and space use CSV: 15,465 records, 31 variables, membership starts in 2021–2023. After removing 15 duplicate rows: 15,450 unique members, with one record summarising each member.

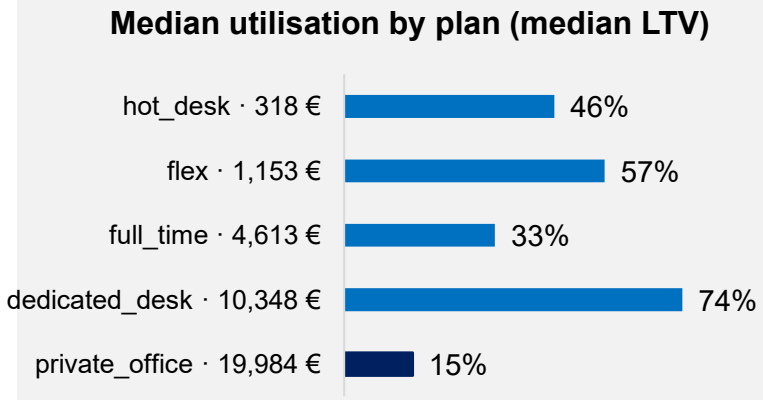
Data area	Available information	Priority variables and contribution to segmentation
Economic value	Total revenue over the membership	lifetime_value, ltv_per_month and ltv_per_visit:Provides the basis for value based segmentation, Lifetime revenue should be interpreted alongside membership duration
Membership	Current plan, entitled days per week, start/end dates and renewals	plan_tier, n_renewals and tenure_months†: Captures differences in access rights and relationship duration
Attendance	Total/weekly visits, last visit and recency, last-period actual/entitled visits	avg_weekly_visits, utilisation and days_since_last_visit: Shows how often members visit compared with how often their plan allows them to
Additional services	Nine usage flags, meeting-room hours, event-space hours and guest passes	uses_ flags, n_addons and meeting_room_hours: Captures service preferences beyond desk access, similar attendance levels may conceal different service needs
Member context	Age, gender, country, industry, team status and colleagues on the account	Helps describe and interpret clusters, for example by industry or team account status

Combining value and behaviour can reveal members with similar revenue but different service needs

Data Understanding – Exploration Report

Members differ in spending, use of their access and service needs

1 High revenue, low utilisation

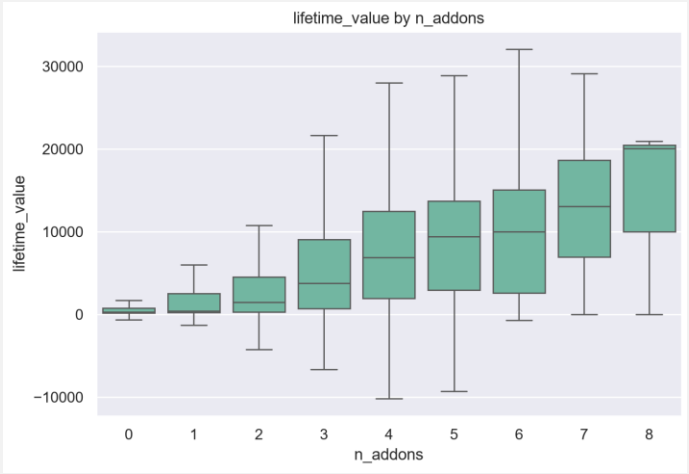


Private office members have the highest median lifetime revenue (€19984), but use only 15% of their visit allowance. Dedicated desk members generate less lifetime revenue (€10348), but use 74%.

For clustering: We have to separate how much members spend from how actively they attend. High revenue alone does not indicate frequent use.

Action: Compare value with actual usage and look for differences within plans

2 Different service needs

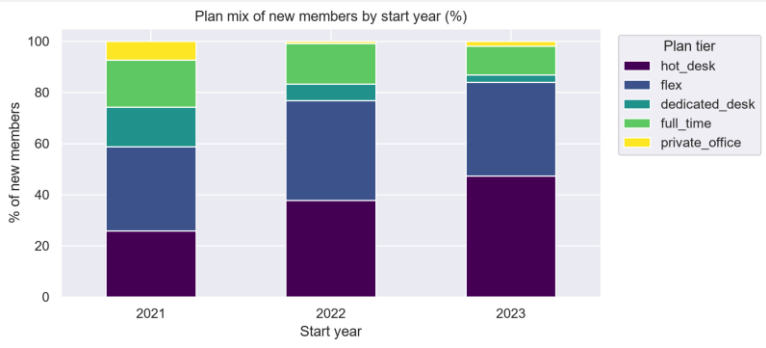


20% of members use no add ons, while 29% use at least three. Median lifetime revenue is €317 among members with no add ons, compared with €9,412 among those using five.

For clustering: We have to identify which services members use to distinguish desk focused members from those using a wider range of services. And also Check whether these differences also reflect their plan or membership duration.

Action: Include service preferences and check plan and tenure effects

3 Different joining-year profiles



Hot desk plans account for 47.3% of members who joined in 2023, compared with 25.8% of those who joined in 2021. These are their recorded current plans, so this does not show which plans they originally chose.

For clustering: Account for membership duration. Newer members have had less time to accumulate revenue and visits and could otherwise be grouped as low-value customers simply because they joined recently

Action: Account for membership duration and don‘t equate new with low value.

Data Understanding – Data Quality

Dimension	Evidence from the ACDC Dataset	Why it matters for clustering	Solutions
completeness	– missing lifetime_value , missed → 1040 → 6,7 % – missing meeting_room_hours , missed →1.295 → 8,4 % – guest_passes_issued, missed → 1269 → 8,2% – Age, missed → 557 → 3,6 % – Gender, missed → 530 → 3,4 %	– Missing values may obscure customer value and service usage patterns, leading to misleading segments – MCAR: Missing rates differ across plans, so MCAR should not be assumed – MAR: Membership plan may help explain missingness, making MAR a plausible working assumption – MNAR: Missingness may also depend on the missing values themselves; this cannot be ruled out	▪ Replace missing values using mean, median, similar observations, or model based estimates alternatively, remove affected records or variables
uniqueness	– 15 fully duplicated member records	– Duplicate observations may slightly overrepresent certain customer profiles and bias cluster formation	▪ Verify member IDS and remove duplicates
validity	– Negative n_renewals: 66 – Negative days_since_last_visit: 56 – Negative total_visits: 42 – Invalid plan_days_per_week: 54	– Invalid values may distort engagement, recency and loyalty measures, potentially misclassifying members	▪ Apply valid range checks, correct impossible values or replace with NaN
Consistency	– 3,954 placeholder membership_end dates (1970-01-01), after deduplication – 55 contradictory meeting-room usage records – Inconsistent plan_tier labels – 94 badge swipes before membership start	– Inconsistent records may create artificial differences in membership tenure, service usage and plan types	▪ Standardise category labels, validate date order and investigate the 1970 placeholder. Cross check service use flags against recorded usage
outliers	– Extreme lifetime_value values – meeting_room_hours up to 41220 – total_visits up to 6530 – avg_weekly_visits up to 47. 4	– A small number of extreme value accounts could dominate value or usage based clustering – ACDC might identify an apparently highly profitable or intensive use segment that primarily reflects exceptional records rather than a meaningful customer group	▪ Check whether extreme values are errors or real customer behavior; correct errors and use robust scaling for valid outliers

Data Preparation – Feature Engeneering

How do the engineered features capture richer customer behaviour?

Feature	How it is built	Evidence from the ACDC dataset	Why it matters for clustering
Utilisation	Actual/entitled visits in the last period	- Median 0.74 (dedicated desk) vs 0.15 (private office) - Almost symmetric (skew –0.04)	Separates members who use their plan from members who only pay for it
n_addons	Number of add-ons used (nine services)	- Median lifetime value: 317 with none, 20,047 with eight - Skew 0.77	Summarises service breadth. Individual flags preserve service preferences and may also support clustering.
tenure_months	Months from start to end (or extraction date)	- Spearman 0.71 with lifetime value - 1% missing	Measures membership duration and helps identify whether higher totals reflect longer membership rather than more intensive behaviour. Assumes that the end date 1970-01-01 means still active
ltv_per_month	Lifetime value / tenure_months	- Median 129 (hot desk) to 698 (private office) - Skew 20.4, 7.6% missing, 79 negative	Average revenue per membership month. Reduces the influence of membership duration, but can be unstable for very short memberships
ltv_per_visit	Lifetime value / total_visits	- Median about 27 - Skew 66.6, 7.3% missing, 51 negative	Value of each visit:separates heavy users from high spenders

Checked: Engineered features: $\rho < 0.90$. Original visit variables: $\rho = 0.93$. visits_per_month excluded ($\rho = 0.97$ with avg_weekly_visits)
Next step (D2): Treat extreme and negative LTV ratios before clustering. Validate membership dates and check whether very short memberships produce extreme monthly revenue values

Hypothesis / Expected Costumer Segments

01 Flexible access users

HOT DESK / FLEX

€129–197 / month · 46–57% use

- Moderate utilization (46–57%) and relatively low monthly revenue (€129–197)
- Members prioritize flexibility over having a permanent workspace
- Moderate usage suggests that these customers may only need a workspace on certain days
- Their relatively low spending and similar usage levels could indicate a shared Customer Segments

02 Highly engaged desk users

DEDICATED DESK

74% use · €394 / month

- Highest utilization (74%) with monthly revenue of approximately €394
- Highest utilization among all membership types
- Regular visits suggest that the coworking space plays an important role in their daily work
- Despite paying less than Private Office members, they make much greater use of their available workspace

03 Premium capacity buyers

PRIVATE OFFICE

€698 / month · 15% use

- Highest monthly revenue (€698), but lowest utilization (15%)
- These customers may value having a private office available whenever they need it, rather than using it every day
- This combination of high spending and low attendance could make them a distinct customer group

04 Lower intensity full time users

FULL TIME

33% use · €305 / month

- Full-time access, but only around one third of available visits are used
- This suggests that customers may not need the workspace as often as their membership allows
- Their usage patterns could be similar to those of other less active members, even with different membership types

Working hypothesis: 3-5 clusters, illustrated by four profiles. These profiles are based on plan level medians. In D2, we will test whether differences in utilisation, monthly revenue and service preferences reveal distinct groups within the same plan

References

1. Aquino, J., & Jonker, A. (n.d.). *Data quality dimensions defined*. IBM. https://www.ibm.com/think/topics/data-quality-dimensions

2. Wirth, R., & Hipp, J. (2000). CRISP-DM: Towards a standard process model for data mining. In *Proceedings of the 4th International Conference on the Practical Applications of Knowledge Discovery and Data Mining* (pp. 29–39). https://www.cs.unibo.it/~danilo.montesi/CBD/Beatriz/10.1.1.198.5133.pdf

3. Dolnicar, S., Grün, B., & Leisch, F. (2018). *Market segmentation analysis: Understanding it, doing it, and making it useful*. Springer. https://doi.org/10.1007/978-981-10-8818-6